

ARCHITECTURE DIAGRAMS

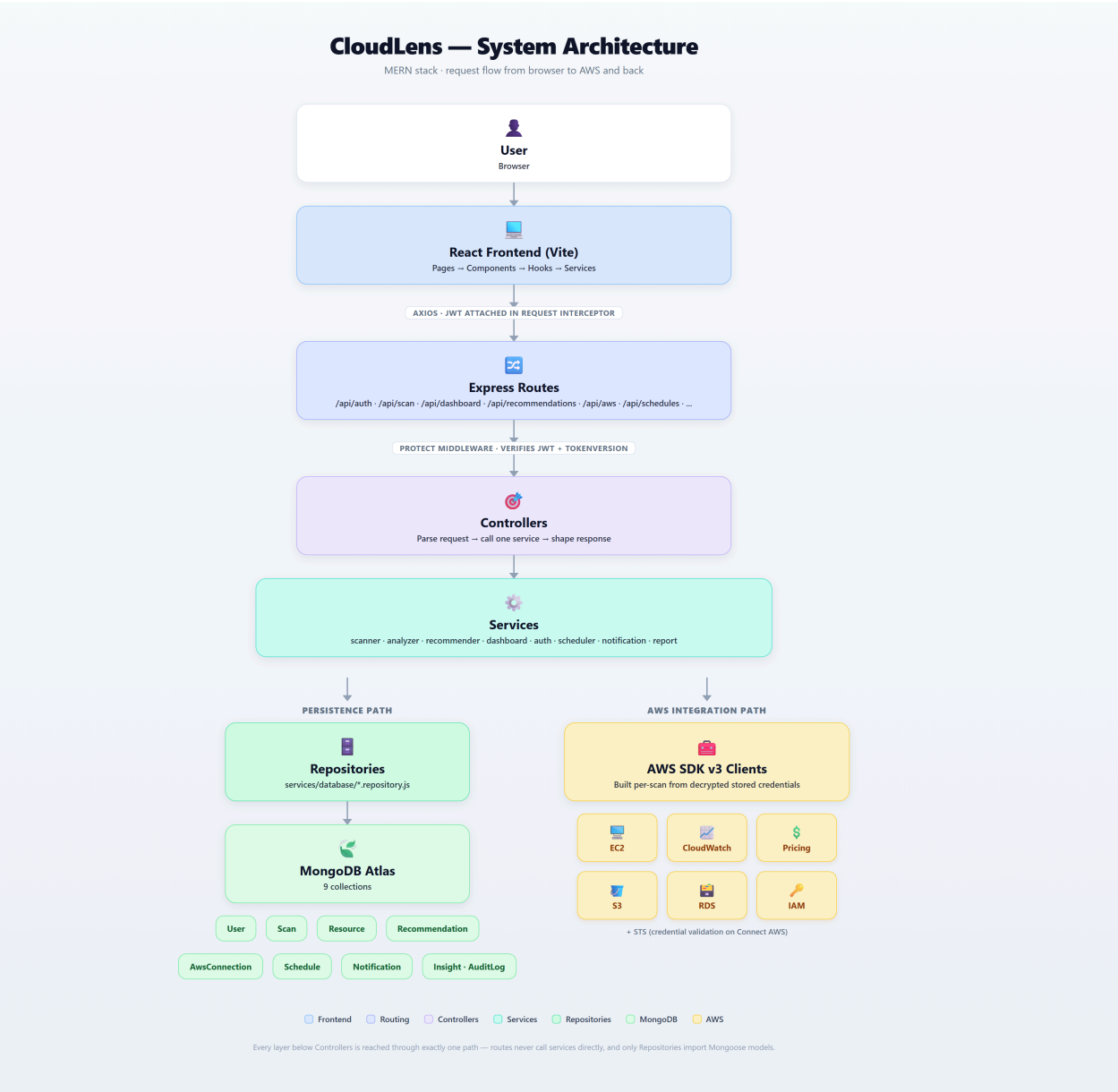
CloudLens

System Architecture & Data Flow Reference

System Architecture · Folder Structure · API Flow · Scan Flow · Recommendation Engine

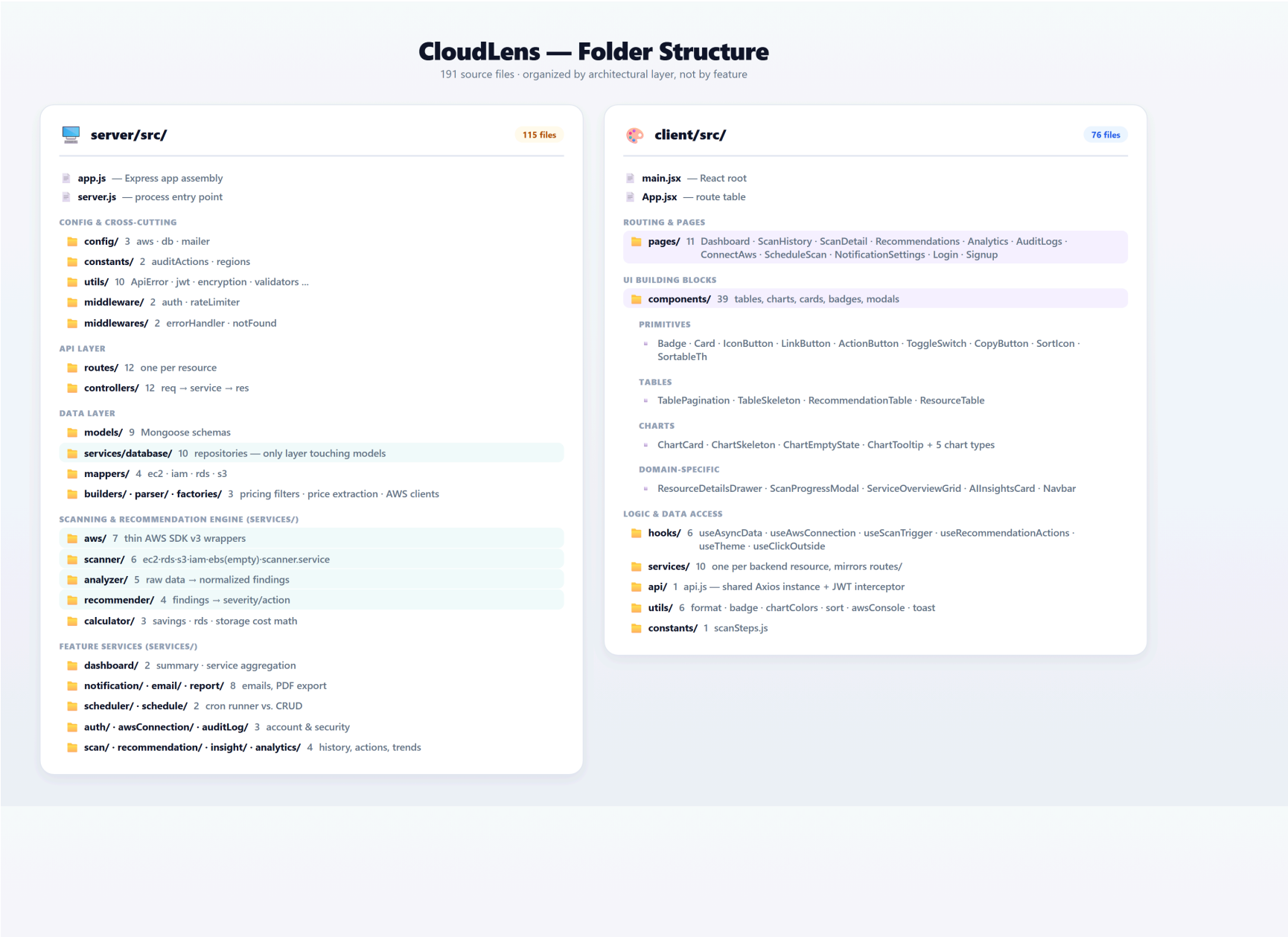
Generated August 1, 2026

1 System Architecture



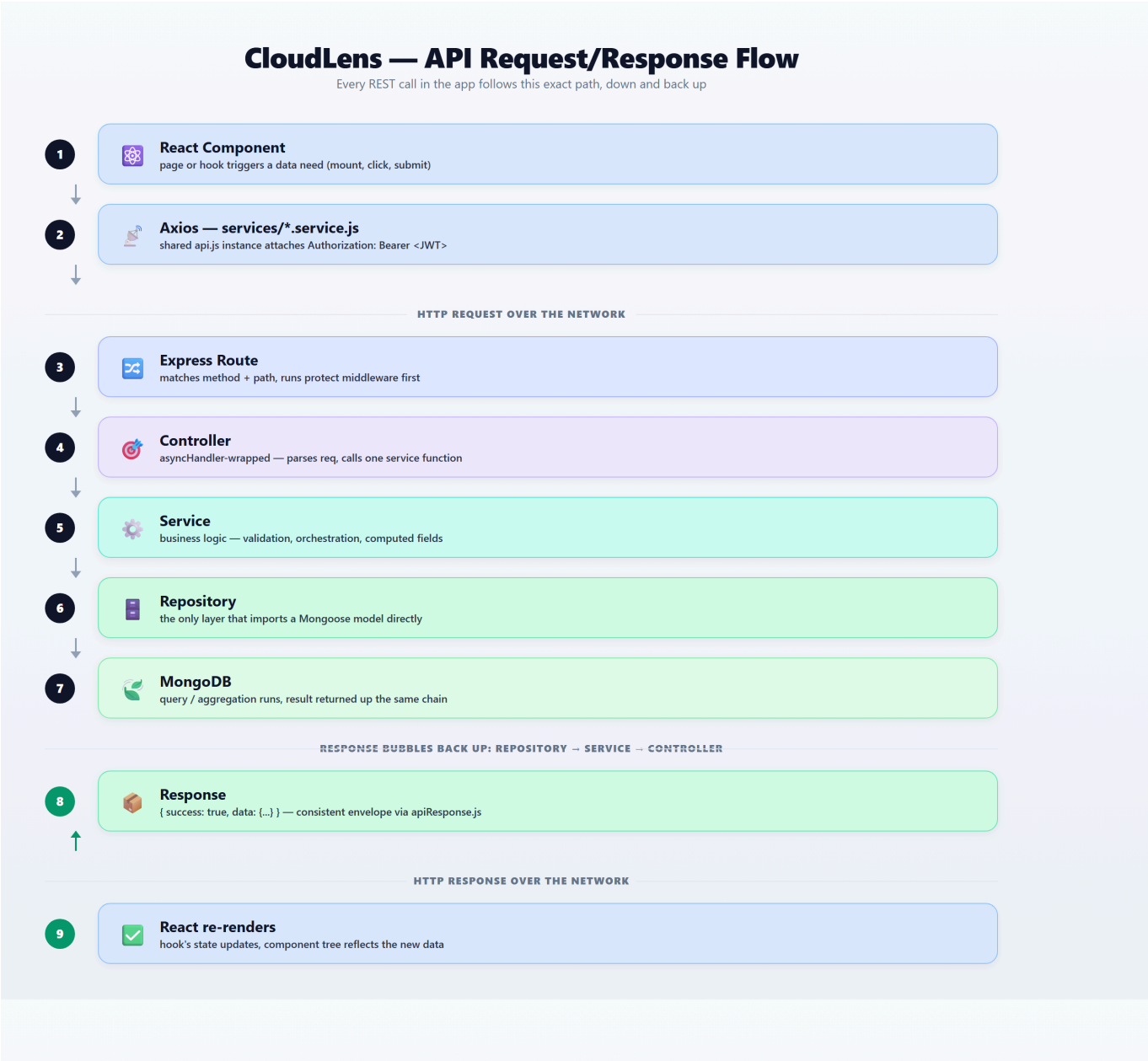
User → React Frontend → Axios (JWT attached) → Express Routes → protect middleware → Controllers → Services, splitting into the Persistence path (Repositories → MongoDB Atlas) and the AWS Integration path (AWS SDK v3 clients → EC2 / CloudWatch / Pricing / S3 / RDS / IAM / STS).

2 Folder Structure



191 source files across server/src (115) and client/src (76), organized by architectural layer rather than by feature — every scanned AWS service gets one file per concern (aws/, scanner/, analyzer/, recommender/), not one folder per service.

3 API Flow



Every REST endpoint in the app follows this exact nine-stage path down and back up. Only the Repository stage is allowed to import a Mongoose model directly — every layer above it goes through exactly one path to reach the database.

4 Scan Flow



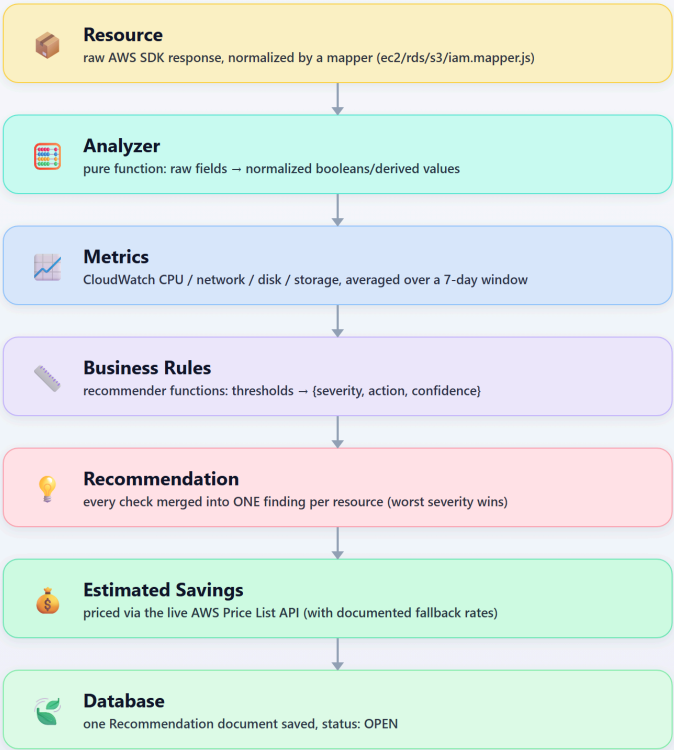
One POST /api/scan fans out to four independent per-service scanners. EC2 runs blocking (a failure fails the whole scan); S3, RDS, and IAM are each individually fault-tolerant, degrading to zero contribution rather than aborting the scan.

5

Recommendation Engine Flow

CloudLens — Recommendation Engine Flow

Per resource, per check — how a raw AWS API response becomes a priced, actionable finding



WORKED EXAMPLE — EC2, STOPPED INSTANCE

Resource: `state: "stopped"` → Analyzer: `isRunning = false` → Metrics: `averageCPU = 0` (no CloudWatch data) → Business Rules: `isRunning` guard short-circuits to `severity: LOW, action: NONE` → Recommendation: "already stopped, no active cost" → Estimated Savings: `$0` → saved as OPEN, LOW severity.

Per resource, per check: raw AWS data → normalized by an analyzer → combined with CloudWatch metrics → evaluated against threshold-based business rules → merged into one recommendation (worst severity wins) → priced → saved as an OPEN Recommendation document.